

Siddhartha Reddy Munugala

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PROFESSIONAL SUMMARY

Computer Science student and full-stack software engineer with hands-on experience building and shipping AI-powered products, real-time web applications, and machine learning pipelines. Founded two live platforms with a combined 1,621 registered users and 20 percent active retention. Currently developing EKO, a proprietary Artificial Intelligence (AI) audio platform with 821 active users and 40 percent daily active user rate at zero paid acquisition cost. Comfortable owning full-stack architecture, cloud deployment, production debugging, and rapid product iteration in fast-paced startup environments.

SKILLS

Programming Languages: Python, TypeScript, JavaScript, SQL

Frameworks and Libraries: Next.js 14, React, Node.js, FastAPI, Tailwind CSS

Artificial Intelligence and Machine Learning: Large Language Models (LLMs), XGBoost, SHAP, PyTorch, Claude Application Programming Interface (API), Gemini API, OpenAI API, LangChain, Whisper,

Retrieval-Augmented Generation (RAG), prompt engineering, agentic workflows

Infrastructure and DevOps: Supabase, PostgreSQL, Neon Postgres, Redis, Docker, Vercel, Render, Git, GitHub Actions, Continuous Integration and Continuous Deployment (CI/CD)

Audio and Media: FFmpeg, Demucs, faster-whisper, librosa, stem separation, waveform processing

Other Technologies: WebRTC, WebSockets, REST APIs, CodeMirror 6, tree-sitter, NetworkX, Google Apps Script

WORK EXPERIENCE

Co-Founder and Lead Engineer — VibeIn and ArenaHQ

January 2025 – Present

vibein.us • arenahq.us

- Built and shipped two full-stack web applications using Next.js 14, Supabase, WebRTC, Twilio, and Gemini Artificial Intelligence, serving 1,621 combined registered users with 20 percent active retention pre-launch.
- Resolved 8 or more critical production incidents including Row-Level Security (RLS) authentication failures, Gemini API deprecation, broken Short Message Service (SMS) and WhatsApp notifications, and Supabase Realtime failures under hard deployment deadlines.
- Implemented row-level security policies, real-time chat, peer recommendation matrices, and peer-to-peer (P2P) camera authentication for live exam proctoring.
- Conducted 50 or more user interviews to drive feature iteration and managed end-to-end Vercel deployment pipelines maintaining 99 percent or higher uptime across both platforms.

Production Operations Staff — USD Athletics, University of South Dakota

August 2025 – Present

Part-Time On-Campus

- Operate live-event audio systems, stadium display boards, and broadcast cues for National Collegiate Athletic Association (NCAA) athletics events under zero-tolerance real-time production constraints.
- Coordinate multi-channel equipment and timing-critical broadcast execution for live audiences with zero margin for error.

Information Technology Coordinator and Publicity Chair — NSLS Executive Board, University of South Dakota

September 2024 – Present

- Built member tracking pipelines, communication scripts, and recruitment dashboards using Python and Google Apps Script, reducing onboarding latency by 50 percent.
- Managed digital member directories and information technology infrastructure for a national collegiate leadership organization.

Freelance Brand and Content Consultant — Independent

June 2019 – December 2022

Remote

- Managed brand identities and content distribution campaigns for 10 or more internet creators and small businesses, maintaining 100 percent client retention across more than 3 years.
- Coordinated asset logistics and brand channel operations under strict non-disclosure agreements (NDAs) and concurrent client deadlines.

PROJECTS

EKO – Artificial Intelligence Background Music and Voice Dubbing System

2026 – Present

- Built a three-mode Artificial Intelligence audio platform: Studio (vocal sketch to 10 Background Music (BGM) variations via Google Lyria 3), BGM Dub (scene-energy-matched music generation for video), and Voice Dub (multilingual speech replacement with background music preservation via Demucs stem separation).
- Resolved critical memory constraint on 2 gigabyte Render container by migrating from local MusicGen-small (1.8 gigabytes RAM) to HuggingFace Inference API, reducing footprint to approximately 400 megabytes; diagnosed and fixed silent loop failures causing single-variation output bug.
- 821 registered users, 40 percent daily active user rate at zero paid acquisition cost. ArXiv preprint (cs.SD) in progress covering full system architecture and unit economics. Stack: Next.js 14, FastAPI, Neon Postgres, Vercel, Render, Gemini Flash, Demucs, librosa, faster-whisper.

PRISM – Pull Request Risk Intelligence and Structural Modeling

2026

- Engineered a machine learning pipeline predicting post-merge engineering risk from 45 features including Abstract Syntax Tree (AST) complexity via tree-sitter, dependency graphs via NetworkX, and code churn metrics.
- Implemented XGBoost classifier with SHapley Additive exPlanations (SHAP) explainability; validated at Area Under the Curve (AUC) score of 0.80 or higher with cross-repository generalization gap below 0.05.
- Authored two formal research proposals with full mathematical appendices for USD faculty and external mentorship review.

COGN – Cognitive Integrity Layer for Artificial Intelligence Codebases

2026

- Built an open-source Continuous Integration and Deployment (CI/CD) GitHub Action that scores pull requests for cognitive drift, which refers to measurable reasoning-pattern shifts introduced by AI-generated code, against repository baselines using three custom primitives.

ALASKA – Encrypted Artificial Intelligence Assistant for Military Veterans

2026 | Top-10 Hackathon Finish

- Built in 48 hours: Advanced Encryption Standard 256-bit Galois/Counter Mode (AES-256-GCM) encrypted Large Language Model (LLM) assistant with Veterans Affairs (VA) routing algorithm, crisis detection, and accessibility modes. Placed in the top 10 at the hackathon.

PETRI-Verse – Autonomous Multi-Agent Artificial Intelligence Simulation

2025

- Concurrent 62-agent Large Language Model civilization simulation with a tick-based event loop; produced an original Artificial Intelligence safety finding around emergent self_modify behavior patterns across agents.

WTFcode – Natural Language Developer Command Line Interface

2026

- Multi-agent agentic developer tool converting natural language instructions to structured multi-step reasoning and code execution pipelines; reduced token overhead by 90 percent via directory import-map compression.

EDUCATION

University of South Dakota

Vermillion, South Dakota • Expected May 2027

Bachelor of Science in Computer Science, Minor in Innovation and Entrepreneurship

Relevant Coursework: Data Structures, Algorithms, Systems Programming, Database Systems, Software Engineering

LEADERSHIP AND ACTIVITIES

- Artificial Intelligence and Entrepreneurship Workshop Facilitator (Self-Initiated, Offline) – Lead community sessions for teenagers ages 13 to 16 covering how Artificial Intelligence integrates into daily life, business fundamentals, customer discovery, product building, and public speaking skills.
- USD INFORMS, Beacom School of Business – Authored 10-week strategic operational plan and built chapter admin portal.
- National Society of Leadership and Success (NSLS) Executive Board – Information Technology Coordinator and Publicity Chair.
- USD Marketing Club – Member.